Logistic Regression

Our recent focus has been on regression problems, where the goal is to predict continuous values, such as "total ice cream sales". In such problems, the output y can take on any real value within some range.

Today, we shift our focus to **classification**, where instead of predicting a continuous value, we aim to predict a discrete category, or class to which an input belongs. In **binary classification**, there are exactly two possible outcomes, typically described as 1 and 0, or positive and negative, but which can be interpreted as yes/no, pass/fail, or accept/reject, etc. There are myriad real-world binary classification problems:

- In email spam detection, the goal is to classify each incoming email as either "spam" or "not spam".
- In medical diagnosis, the goal is to determine whether a patient has a particular disease or not, based on their test results and observed symptoms.
- In image recognition, we might classify images as containing either a cat or a dog.
- For credit card or loan applications, a bank needs to decide whether to approve or deny each application.

The goal of binary classification is to learn a **decision boundary**, which in **linear classification** is a line in a two-dimensional feature space, or a hyperplane in higher dimensions, that separates the data into two distinct classes. (See Figure 1.) Note that kNN and decision trees are *not* linear classifiers.

In **multiclass classification**, there are multiple classes, possibly non-overlapping, such as might arise when labeling regions on a map as urban, rural, mountain, forest, river, lake, etc; or, when labeling activities such as sleeping, walking, running, sitting, etc. from biometric data. In **topic modeling**, however—such as labeling newspaper articles as politics or sports or economics; or labeling movies as drama, comedy, action, horror, etc.—a single data point may be assigned multiple labels.

Why Not Use Linear Regression for Classification? Consider a binary classification problem where our target $y \in \{0, 1\}$. We might be tempted to use a familiar linear model: $p(\boldsymbol{x}) = \boldsymbol{w} \cdot \boldsymbol{x}$. But this approach is less than ideal, for several reasons.

First of all, linear regression can generate predictions that are less than 0 or greater than 1. In a binary classification problem, where we want to predict a 0 or a 1, or interpret our predictions as probabilities, values like $-1/2$ or 3.14 are nonsensical. On the contrary, predictions should lie in the range $[0, 1]$.

Second, standard linear regression loss functions, like root mean squared error, or even mean absolute error, penalize deviations linearly or quadratically, which does not align well with the right-or-wrong nature of binary classification predictions. Number of misclassifications is preferable as an error metric.

Third, linear regression with squared residual loss is sensitive to outliers. Imagine we're classifying tumors as malignant (1) or benign (0) based on tumor size. If we have even a few training examples with very large tumors, the typical linear regression loss lends these outliers disproportionate influence, which can tilt the linear decision boundary, and thereby hurt performance on typical examples. (See Figure 2.)

Finally, linear regression tends to prioritize the majority class, because their errors dominate the total loss.

[1] This notes were compiled in conjunction with Professor Eric Ewing.

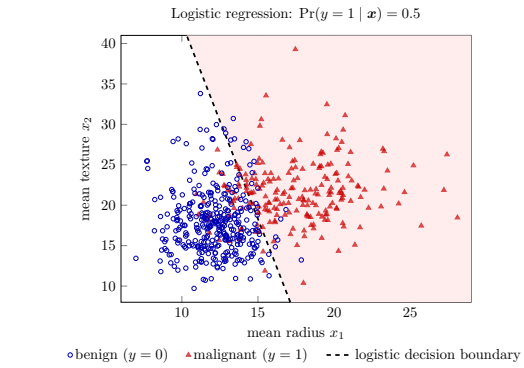


Figure 1: Logistic regression decision boundary on breast cancer data. Features are mean cell radius and mean texture of tumors. Labels are malignant or benign. Predict benign when $z = \boldsymbol{w} \cdot \boldsymbol{x} < 0$; predict malignant when $z = \boldsymbol{w} \cdot \boldsymbol{x} > 0$. Additionally, predict malignant with probability $\hat{y} = \sigma(z) = \sigma(\boldsymbol{w} \cdot \boldsymbol{x})$.

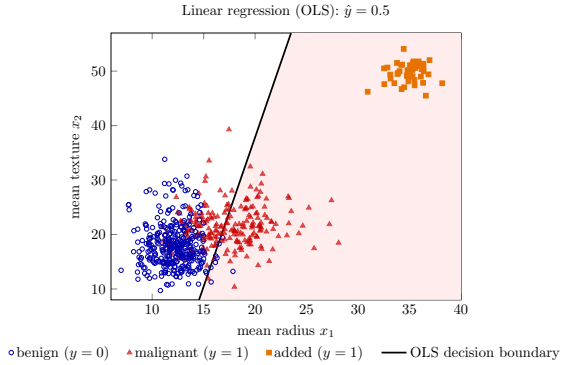


Figure 2: Linear regression decision boundary on breast cancer data with artificial outliers. Predict benign when $\hat{y} = \boldsymbol{w} \cdot \boldsymbol{x} < 0.5$; predict malignant when $\hat{y} = \boldsymbol{w} \cdot \boldsymbol{x} > 0.5$.

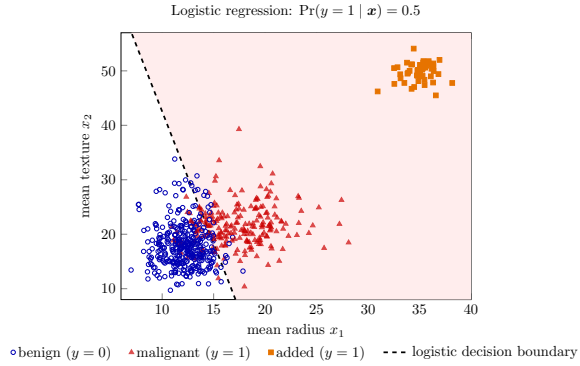


Figure 3: Logistic regression decision boundary on breast cancer data with artificial outliers. Predict benign when $z = \boldsymbol{w} \cdot \boldsymbol{x} < 0$; predict malignant when $z = \boldsymbol{w} \cdot \boldsymbol{x} > 0$. Additionally, predict malignant with probability $\hat{y} = \sigma(z) = \sigma(\boldsymbol{w} \cdot \boldsymbol{x})$.

1 Logistic Regression

What we will do instead is build a linear classifier using a function that takes any real-valued input and squashes it into the range $[0, 1]$, yielding valid probabilities. This supervised learning technique is called **logistic regression**, and the squashing function it relies on is called the **sigmoid** function.

1.1 The Sigmoid Function

The **sigmoid** function (see Figure 4), also called the **logistic** function, is defined as follows: for $z \in \mathbb{R}$,

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The most important property of the sigmoid function is its range. No matter what value z we input, it always outputs a value strictly between 0 and 1, approaching these bounds asymptotically.

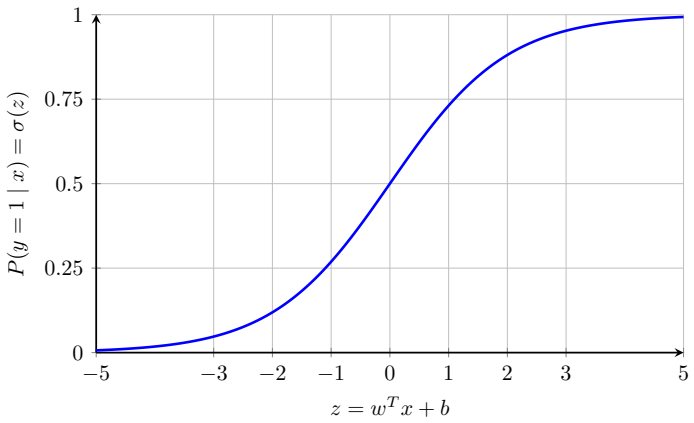


Figure 4: The sigmoid function

1.2 The Logistic Regression Model

We can now define the logistic regression model. It applies the sigmoid to the familiar linear combination:

$$p(\boldsymbol{x}) \doteq p(\boldsymbol{x}; \boldsymbol{w}) = \sigma(\boldsymbol{w} \cdot \boldsymbol{x}) = \frac{1}{1 + e^{-\boldsymbol{w} \cdot \boldsymbol{x}}}$$

We then interpret $p(\boldsymbol{x})$ to mean the probability $Y = 1$ given the input x and parameters $\boldsymbol{w}$. In other words,

$$p(\boldsymbol{x}) = \Pr(Y = 1 \mid \boldsymbol{x}; \boldsymbol{w})$$

Since there are only two classes in binary classification, this interpretation implies that

$$\Pr(Y = 0 \mid \boldsymbol{x}; \boldsymbol{w}) = 1 - p(\boldsymbol{x})$$

For example, if $p(\boldsymbol{x}) = 0.75$, we interpret this prediction as saying there is a 75% chance this data point belongs to the positive class and a 25% chance it belongs to the negative class.

1.3 Decision Boundary

In practice, we usually need to make a hard decision: should we classify an example as positive or negative? (E.g., tumor / surgery or not? approve loan application or not?) The standard approach is to predict 1 if the probability estimate is at least 0.5, and predict 0, otherwise. That is, we predict 1 whenever:

$$p(\boldsymbol{x}) \geq 0.5$$

Because the sigmoid outputs 0.5 on input 0, this is equivalent to defining the **decision boundary** as:

$$\boldsymbol{w} \cdot \boldsymbol{x} \geq 0$$

For example, in two dimensions, with features x_1 and x_2, the decision boundary is the line:

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

Despite classifying based on the nonlinear sigmoid function, logistic regression is a linear classifier because its decision boundary is linear. Points on one side of this line are classified as positive, while points on the other side are classified as negative. Moreover, the further a point lies from the boundary, the more confident the model is in its classification, as reflected by predictions / probabilities closer to 0 or 1.

1.4 Properties of the Sigmoid

In addition to its range, the sigmoid function is well suited to our purposes for a few additional reasons.

- It is monotonically increasing in its input; that is, bigger inputs always produce bigger outputs. Thus, it preserves order in the linear combination $\boldsymbol{w} \cdot \boldsymbol{x}$. So, if a point is more likely than another to be in the positive class based on its raw linear combination, the sigmoid function preserves this ordering.
- It is symmetric around the point $(0, 0.5)$: i.e., $\sigma(-z) = 1 - \sigma(z)$. When the linear combination $\boldsymbol{w} \cdot \boldsymbol{x} = 0$, the sigmoid outputs exactly 0.5, which we interpret as maximal uncertainty between the two classes.
- The sigmoid is smooth and differentiable everywhere, which is crucial for gradient-based optimization. Moreover, as you'll see next, its derivative takes a particularly simple form.

1.5 Derivative of the Sigmoid

Let's derive the derivative of the sigmoid:

$$\frac{d\sigma(z)}{dz} = \frac{d}{dz}\left(\frac{1}{1+e^{-z}}\right)$$

We rewrite this derivative as $(1+e^{-z})^{-1}$, and apply the chain rule:

$$\begin{aligned} &= -(1+e^{-z})^{-2} \cdot (-e^{-z}) \\ &= \frac{e^{-z}}{(1+e^{-z})^2} \end{aligned}$$

We then factor this expression, and simplify:

$$\begin{aligned}
&= \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} \\
&= \frac{1}{1+e^{-z}} \cdot \frac{(1+e^{-z})-1}{1+e^{-z}} \\
&= \frac{1}{1+e^{-z}} \cdot \left(1 - \frac{1}{1+e^{-z}}\right) \\
&= \sigma(z) \cdot (1-\sigma(z))
\end{aligned}$$

The derivative of the sigmoid function is thus:

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1-\sigma(z))$$

Notice how simple this is! This is not just a surprising fact; it is also computationally convenient. Once we've computed $\sigma(z)$, we can compute its derivative using just one multiplication and one subtraction, without reevaluating any exponentials.

A The Sigmoid Function and the Log-odds are Inverses

Recall that a Bernoulli random variable Y is a discrete random variable with two possible outcomes, $\{0, 1\}$, and success probability p. The **odds** of a Bernoulli random variable is defined as $p/1-p$. Logistic regression models the **log-odds** (*a.k.a.* the **logit**) as a linear function of the features:

$$\text{logit}\left(p(\boldsymbol{x})\right) = \log\left(\frac{p(\boldsymbol{x})}{1-p(\boldsymbol{x})}\right) = \boldsymbol{w}\cdot\boldsymbol{x} \;\equiv\; z \tag{1}$$

Since $p \in (0,1)$, the odds range from $0 = 0/1$ to $1/(1-1)$, which tends to ∞. Thus, the log-odds ranges over all of $\mathbb{R}$, so making a linear predictor z is a natural fit on this scale.

Relationship with the Sigmoid. To recover p from Equation 1, we exponentiate both sides and solve:

$$\frac{p}{1-p} = e^z \tag{2}$$

$$p = e^z(1-p) \tag{3}$$

$$pe^z + p = e^z \tag{4}$$

$$p\left(1+e^z\right) = e^z \tag{5}$$

$$p = \frac{e^z}{1+e^z} \;=\; \frac{1}{1+e^{-z}}, \tag{6}$$

where the final form is the familiar sigmoid function, and the inverse of the log-odds! (See Figure 5.) Hence,

$$p(\boldsymbol{x}) = \sigma(\boldsymbol{w}\cdot\boldsymbol{x}) \tag{7}$$

In other words, fitting a linear model to the log-odds is equivalent to passing a linear fit through the sigmoid to obtain a probability.

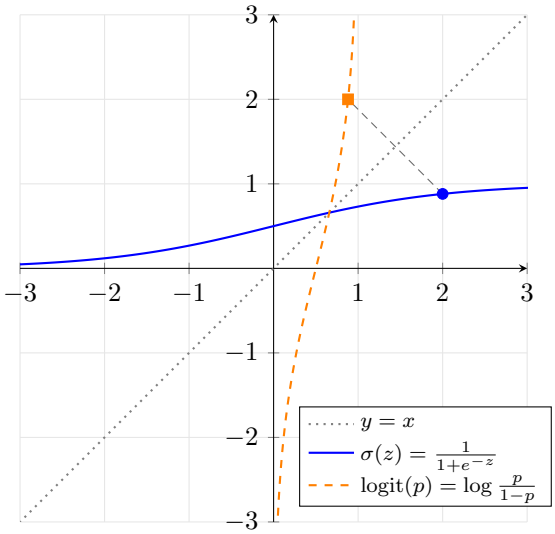


Figure 5: The sigmoid and the log-odds are reflections of one another across $y = x$, i.e., they are inverses.